



Optimized $GF(2^{128})$ Multiplier for AES-GCM

Karatsuba Decomposition and
Microarchitectural Pipelining

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Presentation Outline

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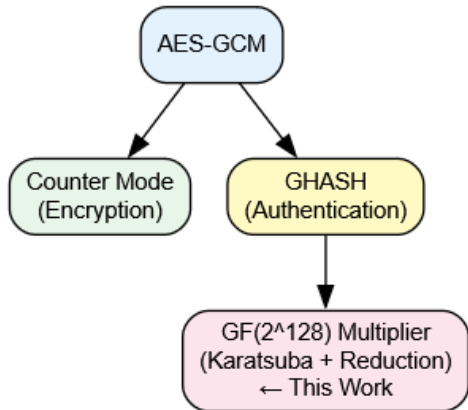
- Introduction
- Problem Formulation & Objectives
- Novelty
- Block Diagram
- System Design
- Implementation
- Results & Tabulation
- SDG Relevance
- Societal & Environmental Impact
- Conclusion
- References



Why Optimize $GF(2^{128})$ Multiplication?

1 Introduction

- AES-GCM widely used in secure communication
- Authentication performed using GHASH
- Core operation: $GF(2^{128})$ multiplication
- Multiplier determines throughput and frequency
- This work optimizes the multiplier architecture





Problem Formulation & Objectives

2 Problem Formulation & Objectives

- $GF(2^{128})$ multiplication is the computational bottleneck in GHASH
- Naive polynomial multiplier has large critical path
- Algorithmic optimization alone gives limited delay improvement
- Need microarchitectural techniques to scale frequency

Objectives:

- Compare naive and Karatsuba-based architectures
- Introduce internal pipelining for delay reduction
- Evaluate area, frequency, power and performance density



Novelty of the Proposed Work

3 Novelty

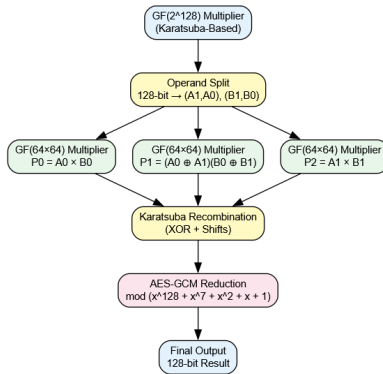
- Implemented a reduced naive $GF(2^{128})$ multiplier as baseline
- Applied **partial Karatsuba decomposition** only at 128-bit level
- No recursive Karatsuba inside $GF(2^{64})$ blocks
- Identified critical path inside $3 \times GF(2^{64})$ multipliers
- Introduced **internal pipelining** within $GF(2^{64})$ structure

Key Insight: Performance improvement achieved through microarchitectural pipelining rather than deeper recursive decomposition.



Partial Karatsuba-Based $GF(2^{128})$ Architecture

4 Block Diagram

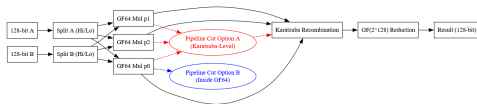




Critical Path Analysis

4 Block Diagram

- Karatsuba reduces multiplication count ($4 \rightarrow 3$)
- Each $GF(2^{64})$ multiplier remains combinational
- Critical path dominated by:
 - 64-bit partial product tree
 - XOR recombination logic
 - Final reduction network
- Frequency improvement is limited without pipelining

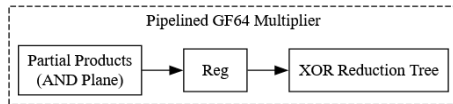




64-bit Polynomial Multiplier Block

5 System Design

- 64×64 polynomial multiplication
- Partial product (AND plane) generation
- XOR reduction tree for accumulation
- Produces 128-bit intermediate result
- Critical path dominated by XOR tree depth





GF(2¹²⁸) Field Representation

5 System Design

Multiplication performed over GF(2¹²⁸) using irreducible polynomial:

$$x^{128} + x^7 + x^2 + x + 1$$

- Binary polynomial arithmetic
- XOR implements addition
- Final 128-bit modular reduction applied after Karatsuba recombination
- Reduction logic contributes to overall critical path



Implementation Overview

6 Implementation

- RTL implemented in Verilog
- Three architectures synthesized:
 - Reduced Naive GF(2^{128})
 - Partial Karatsuba (Combinational)
 - Karatsuba with Internal Pipelining
- Target library: SAED 14nm RVT
- Same synthesis constraints for fair comparison



Reduced Naive $\text{GF}(2^{128})$ Architecture

6 Implementation

- Full 128×128 polynomial multiplication
- AND plane for partial product generation
- XOR tree for accumulation
- Final 128-bit modular reduction
- 12 levels of logic (critical path)
- Fully combinational (1-cycle latency)



Partial Karatsuba Implementation

6 Implementation

- 128-bit inputs split into 64-bit halves
- Karatsuba applied only at top level:
 - 3×64 -bit polynomial multipliers
- No recursive Karatsuba inside 64-bit blocks
- 11 logic levels (slight reduction vs naive)
- Combinational architecture (1-cycle)



Pipelined Karatsuba Implementation

6 Implementation

- Pipeline registers inserted inside 64-bit multiplier blocks
- Logic partitioned into 3 stages:
 - Stage 1: Partial product generation
 - Stage 2: XOR compression
 - Stage 3: Karatsuba recombination + final reduction
- Total latency: 3 cycles



Synthesis Setup

6 Implementation

- Technology: SAED 14nm RVT standard cell library
- Synopsys Design Compiler
- Same timing constraints across architectures
- No manual retiming
- Wireload and operating conditions kept identical



Area & Structural Comparison

7 Results & Tabulation

Metric	Naive	Karatsuba	Pipeline
Total Area (μm^2)	33080	26747	26756
Combinational Cells	23640	17505	19010
Sequential Cells	0	0	1215
Logic Levels	12	11	9

Observation: 19% area reduction using Karatsuba; negligible area overhead due to pipelining.



Timing & Performance Comparison

7 Results & Tabulation

Metric	Naive	Karatsuba	Pipeline
Critical Path (ns)	4.40	4.20	2.23
F_{max} (MHz)	238	250	448
Latency (cycles)	1	1	3
Throughput (Mops/s)	238	250	448

Key Result: 88% increase in maximum frequency.



Power & Efficiency Metrics

7 Results & Tabulation

Metric	Naive	Karatsuba	Pipeline
Dynamic Power (mW)	3.864	3.156	7.816
Leakage (μ W)	425	315	315
Performance/Area (MHz/ μ m ²)	0.00719	0.00934	0.0167
Performance/Watt (MHz/mW)	61.6	79.2	57.3

Insight: Performance density improves by 132% with pipelining.



Architectural Trade-Off Analysis

7 Results & Tabulation

- Karatsuba:
 - Reduces multiplication count ($4 \rightarrow 3$)
 - 19% area reduction
 - Marginal timing improvement
- Internal Pipelining:
 - Breaks long combinational path
 - Reduces critical path from 4.4 ns $\rightarrow$ 2.23 ns
 - 88% increase in F_{max}
 - Latency increases ($1 \rightarrow 3$ cycles)
 - Dynamic power increases due to higher switching

Conclusion: Microarchitectural optimization provides greater frequency scaling than algorithmic optimization alone.



SDG Relevance

8 SDG Relevance

- Secure communication is critical for digital infrastructure
- AES-GCM widely used in:
 - TLS/SSL
 - VPNs
 - Cloud data centers
- Efficient hardware reduces energy per encrypted transaction
- Supports SDG 9: Industry, Innovation and Infrastructure



Societal & Environmental Impact

9 Societal & Environmental Impact

- Higher throughput reduces processing bottlenecks
- Energy-efficient accelerators reduce data center load
- Lower silicon area improves manufacturability
- Enables scalable secure IoT and embedded systems



Conclusion

10 Conclusion

- Evaluated three $GF(2^{128})$ multiplier architectures
- Karatsuba reduced area by 19%
- Internal pipelining reduced critical path by 47%
- Achieved 88% increase in maximum frequency
- Performance density improved by 132%

Final Insight: Balanced algorithmic and microarchitectural optimization yields the most efficient design.



References

11 References

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Thank you